

Deploy an Automatic build pipeline using DockerHub

Step 1 - Initialize a simple Dockerfile

- Run `nano Dockerfile`
- Write a basic dockerfile

```
(~/Desktop/devops/lab6) (1rv24mc038_shreerakshahr@asus:pts/1)
(14:48:08) -> nano Dockerfile (Mon, Dec01)
(~/Desktop/devops/lab6) (1rv24mc038_shreerakshahr@asus:pts/1)
(14:49:12) -> cat Dockerfile (Mon, Dec01)
FROM alpine:latest
CMD ["echo", "Hello world"]
```

Step 2 - Build the image

- Run `docker build -t lab6 .`

```
(~/Desktop/devops/lab6) (1rv24mc038_shreerakshahr@asus:pts/1)
(14:49:18) -> docker build -t lab6 . (Mon, Dec01)
[+] Building 5.5s (5/5) FINISHED docker:default
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 84B 0.0s
=> [internal] load metadata for docker.io/library/alpine:latest 5.4s
=> [internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> CACHED [1/1] FROM docker.io/library/alpine:latest@sha256:4b7ce07002c69e8f3d704a9c5d6fd3053be500b7f1c69fc0d8 0.0s
=> exporting to image 0.0s
=> => exporting layers 0.0s
=> => writing image sha256:68c9844fd9bd5771491705f048281fddf46e9904ee96e0a5e73d8319afb9b4f6 0.0s
=> => naming to docker.io/library/lab6 0.0s
```

Step 3 - Generate a PAT from GitHub

- Go to Settings -> Developer Settings -> Personal Access Token -> Tokens(classic)
- Provide `write:packages` and `read:packages` scope
- Click on generate token and copy the token to a safe place

Step 4 - Login to GHCR using the PAT

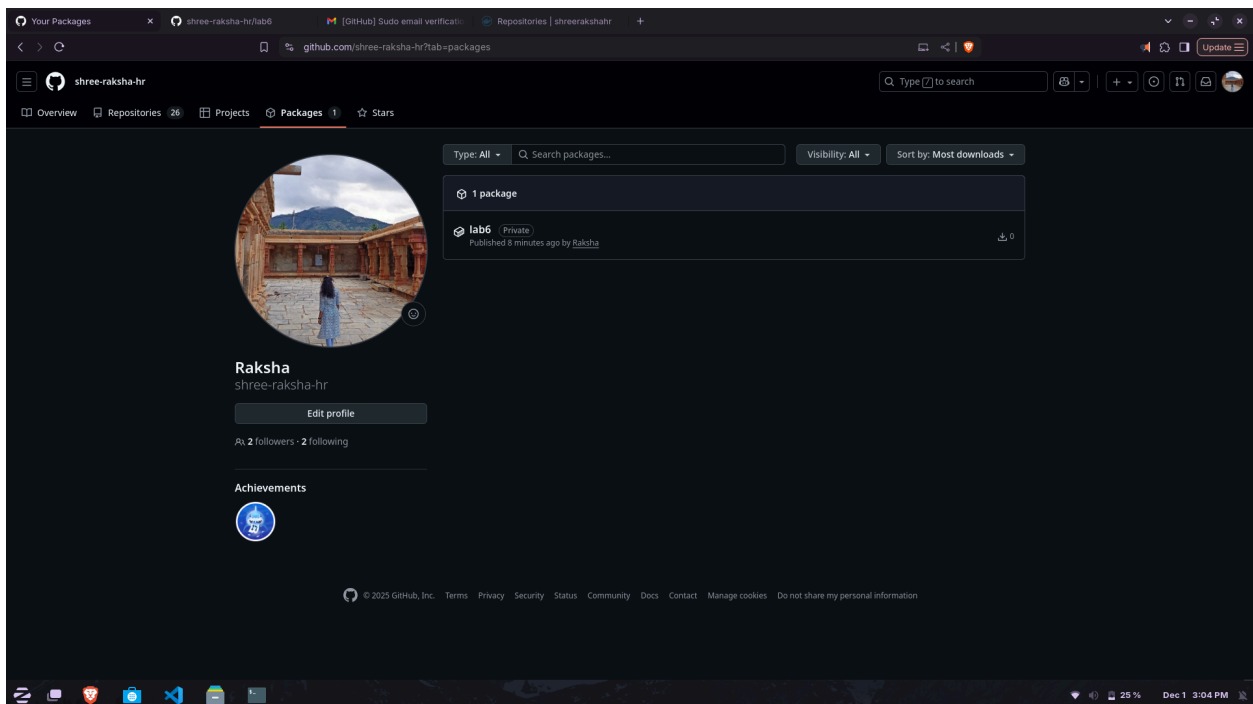
- Run the following command to login

```
(~/Desktop/devops/lab6) (1rv24mc038_shreerakshahr@asus:pts/1)
(14:49:18) -> docker build -t lab6
[+] Building 5.5s (5/5) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 84B
=> [internal] load metadata for docker.io/library/alpine:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> CACHED [1/1] FROM docker.io/library/alpine:latest@sha256:4b7ce07002c69e8f3d704a9c5d6fd3053be500b7f1c69fc0d8
=> exporting to image
=> => exporting layers
=> => writing image sha256:68c9844fd9bd5771491705f048281fddf46e9904ee96e0a5e73d8319afb9b4f6
=> => naming to docker.io/library/lab6
```

Step 5 - Push the image to GHCR

- First tag the image to follow *ghcr.io/<github username>/<image name>*
- Push the docker image
- The image will be present under packages

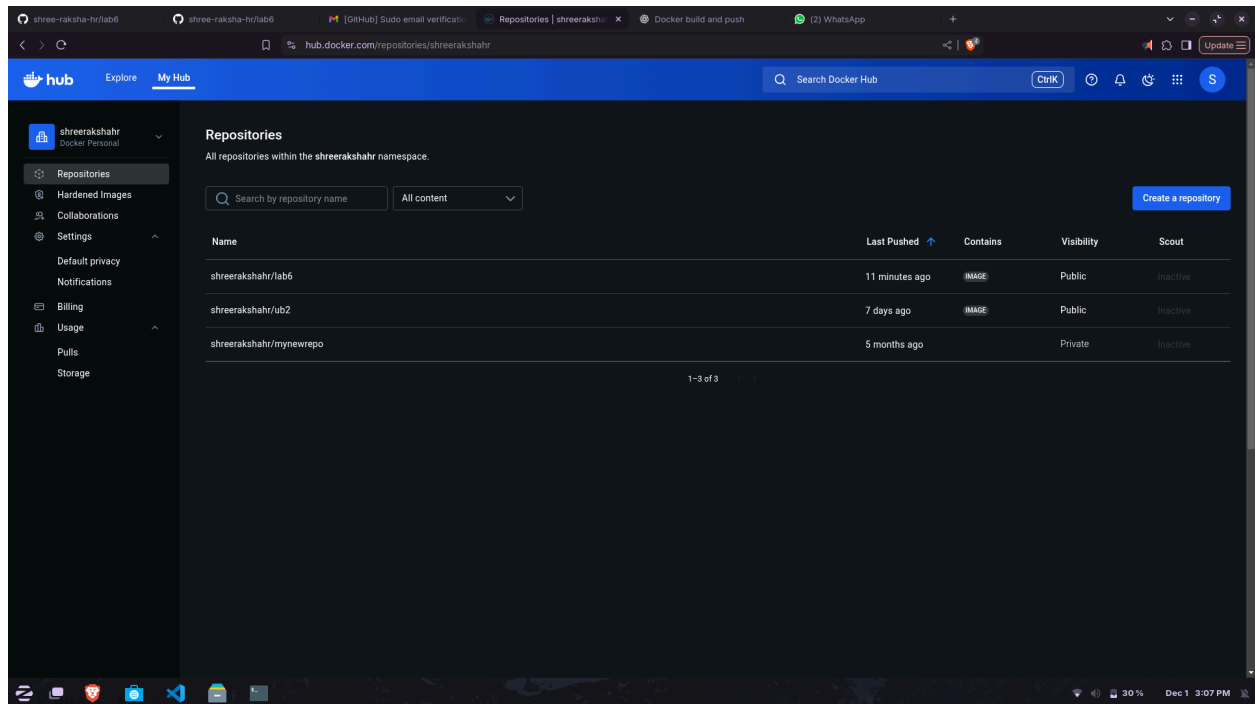
```
(~/Desktop/devops/lab6) (1rv24mc038_shreerakshahr@asus:pts/1)
(14:51:54) -> docker tag lab6 ghcr.io/shree-raksha-hr/lab6
(~/Desktop/devops/lab6) (1rv24mc038_shreerakshahr@asus:pts/1)
(14:52:41) -> docker push ghcr.io/shree-raksha-hr/lab6
Using default tag: latest
The push refers to repository [ghcr.io/shree-raksha-hr/lab6]
256f393e029f: Preparing
unauthorized: unauthorized: User cannot be authenticated with the token provided.
```



Step 6 - Push the image to DockerHub

- First tag the image to follow `<dockerhub username>/<image name>`
- Push the image

```
Error response from daemon: No such image: shreerakshahr/lab6:latest
(~/Desktop/devops/Lab6) (1rv24mc038_shreerakshahr@asus:pts/1)
(14:58:55) → docker tag lab6 shreerakshahr/lab6 1 ↵ (Mon, Dec 01)
(~/Desktop/devops/Lab6) (1rv24mc038_shreerakshahr@asus:pts/1)
(14:55:57) → sudo docker push shreerakshahr/lab6 1 ↵ (Mon, Dec 01)
Using default tag: latest
The push refers to repository [docker.io/shreerakshahr/lab6]
256f393e029f: Mounted from library/alpine
latest: digest: sha256:146add83900056261b2f6e9efae2be2ca202ac1fc96903bfe1f6803ccba2bb8 size: 527
```



Step 7 - Set up git repository

- Run the following commands in the main folder

```
(~/Desktop/devops/lab6) (1rv24mc038_shreerakshahr@asus:pts/1)
(14:56:20) -> git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/raksha/Desktop/devops/lab6/.git/
(~/Desktop/devops/lab6) (1rv24mc038_shreerakshahr@asus:pts/1)
(14:56:36 on master *) -> git add .
(~/Desktop/devops/lab6) (1rv24mc038_shreerakshahr@asus:pts/1)
(14:56:41 on master +) -> git commit -m "initial commit"
[master (root-commit) 515b5c5] initial commit
1 file changed, 2 insertions(+)
create mode 100644 Dockerfile
(~/Desktop/devops/lab6) (1rv24mc038_shreerakshahr@asus:pts/1)
(14:56:47 on master) -> git branch -M main
(~/Desktop/devops/lab6) (1rv24mc038_shreerakshahr@asus:pts/1)
(14:57:49 on main) -> git remote add origin git@github.com:shree-raksha-hr/lab6.git
(~/Desktop/devops/lab6) (1rv24mc038_shreerakshahr@asus:pts/1)
(14:57:56 on main) -> git push -u origin main
```

```
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 263 bytes | 263.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:shree-raksha-hr/lab6.git
 * [new branch]      main -> main
Branch 'main' set up to track remote branch 'main' from 'origin'.
```

